

AMAZON SALES DASHBOARD ANALYSIS

Data Storytelling & Statistical Validation
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objective

The objective of this project is to analyze Amazon sales data, identify customer purchasing trends, evaluate sales performance, and provide business recommendations using data visualization and statistical validation techniques.

INTRODUCTION

This project focuses on analyzing Amazon sales data to understand sales trends, customer behavior, product performance, and profit distribution. Python and Power BI were used to clean, analyze, and visualize the data for business decision-making.

DATA CLEANING & PREPROCESSING

The dataset was cleaned and preprocessed to improve data quality and ensure accurate analysis. Missing values and duplicate records were checked, data consistency was verified, and the cleaned dataset was prepared for visualization and business analysis.

DATASET OVERVIEW

Data Dictionary: Amazon Sales Dataset			
Column Name	Data Type	Description	Business Relevance
product_id	String	Unique identifier for each product.	Tracking individual SKU performance.
product_name	String	Full title of the product.	Identifying what is being sold.
category	String	Hierarchy of categories (e.g., Electronics Accessories).	Analyzing sales by department.
discounted_price	Float	The price after discount (in ₹).	Primary revenue metric.
actual_price	Float	The original retail price (MSRP).	Calculating total potential value.
discount_percentage	Float	Percentage off the actual price.	Evaluating the effectiveness of promos.
rating	Float	Average customer star rating (1-5).	Measuring customer satisfaction.
rating_count	Integer	Total number of users who rated the product.	Determining the reliability of the rating.
about_product	String	Detailed description of the product features.	Useful for text mining/NLP.

The dataset contains detailed product, pricing, discount, and customer review information used for sales and business analysis.

DATA CLEANING & PREPROCESSING

```
product_id      0
product_name    0
category        0
discounted_price 0
actual_price    0
discount_percentage 0
rating          0
rating_count    2
about_product   0
user_id         0
user_name       0
review_id       0
review_title    0
review_content  0
img_link        0
product_link    0
```

```
print("Duplicate Rows:", df.duplicated().sum())
```

Duplicate Rows: 0

```
df = df.drop_duplicates()
print("Duplicates Removed ✓")
```

Duplicates Removed ✓

```
print("File Saved as cleaned_amazon.csv ✓")
```

... File Saved as cleaned_amazon.csv ✓

```
print(df.isnull().sum())
```

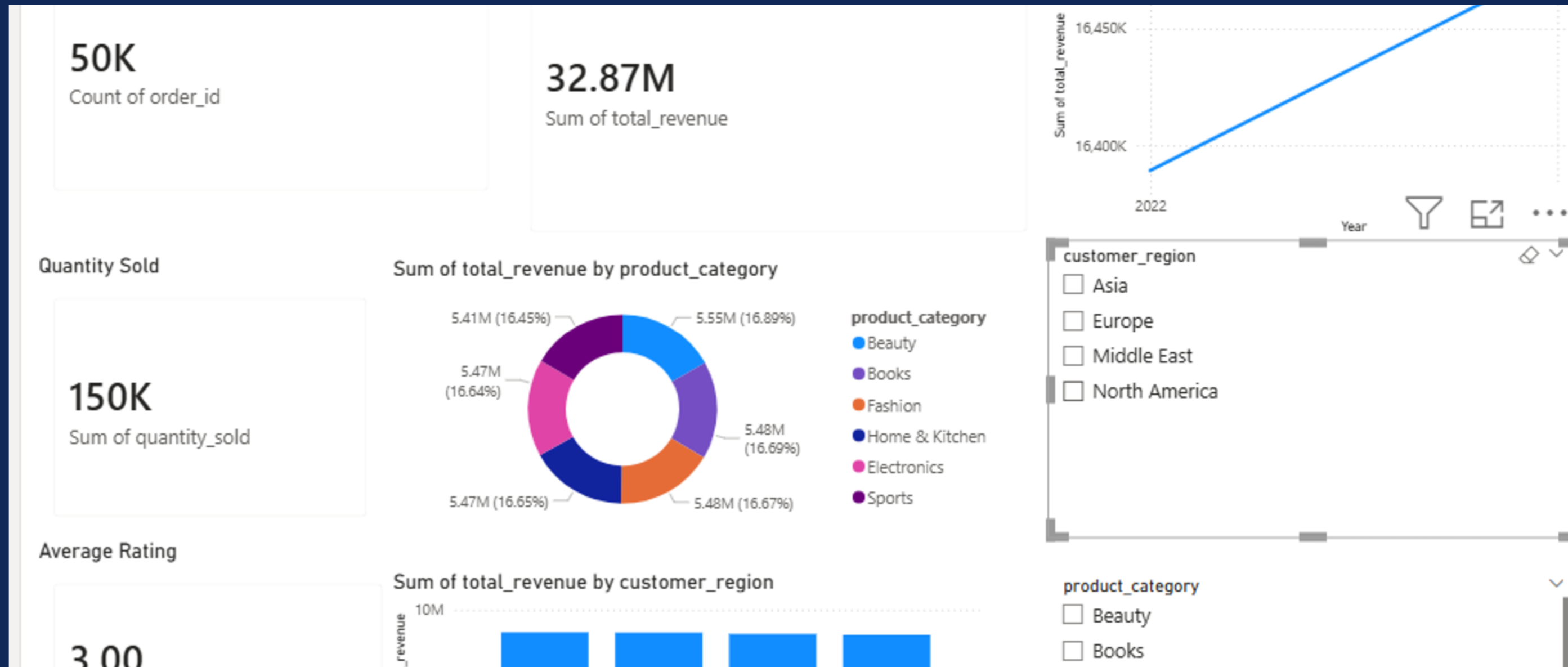
```
product_id      0
product_name    0
category        0
discounted_price 0
actual_price    0
discount_percentage 0
rating          0
rating_count    0
about_product   0
user_id         0
user_name       0
review_id       0
review_title    0
review_content  0
img_link        0
product_link    0
price_difference 0
```

Data preprocessing included handling missing values, removing duplicate records, validating data consistency, and preparing the dataset for analysis.

POWER BI DASHBOARD

The Power BI dashboard provides interactive visualization of sales performance, customer ratings, discounts, profit trends, and category-wise analysis for better business decision-making.

POWER BI DASHBOARD



The Power BI dashboard provides interactive visualization of sales performance, revenue trends, product categories, customer regions, and ratings to support business analysis and decision-making.

SALES INSIGHTS

- Total revenue reached 32.87M from overall sales.
- More than 50K customer orders were recorded.
- Total quantity sold exceeded 150K products.
- Multiple product categories contributed significantly to overall revenue.
- Monthly revenue trends showed steady business performance.
- Different customer regions contributed significantly to sales growth.

CUSTOMER & PROFIT INSIGHTS

- Average customer rating was 3.0
- Product ratings influenced customer purchases
- Different regions showed varying revenue performance
- Discounts impacted product sales
- Customer feedback helped evaluate product quality

HYPOTHESIS TESTING

- **Objective** : To determine whether discounts significantly affect product sales.
- **Null Hypothesis (H_0)** : Discounts do not significantly affect sales performance.
- **Alternative Hypothesis (H_1)** : Discounts significantly affect sales performance.
- **Statistical Rule:**

CONCLUSION

The Amazon Sales Dashboard Analysis project successfully analyzed sales, customer behavior, product categories, and revenue trends using Python and Power BI. Data cleaning and preprocessing improved dataset quality, while interactive dashboards provided meaningful business insights. Statistical validation showed that discount strategies significantly influenced sales performance. Overall, the project demonstrated how data-driven analysis can support better business decision-making and improve sales strategies.



THANK YOU

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